

Automated EDA: Pandas Profiling

Module 22 Study Notes • Accelerating Exploratory Data Analysis

1. What is Pandas Profiling?

In the previous modules, we performed Exploratory Data Analysis (EDA) manually by writing separate code blocks for Univariate Analysis, Bivariate Analysis, correlation heatmaps, and missing value checks.

Pandas Profiling (now widely known as *ydata-profiling*) is an open-source Python library that automates this entire process.

With just a few lines of code, it generates a comprehensive, interactive HTML report containing all standard EDA metrics, saving Data Scientists hours of manual plotting.

2. Installation and Basic Usage

First, install the library using your terminal or command prompt:

```
pip install pandas-profiling
```

Next, generate the report inside your Jupyter Notebook:

```
from pandas_profiling import ProfileReport

# Generate the report object
prof = ProfileReport(df)

# Export it as an interactive HTML file
prof.to_file(output_file='output.html')
```

3. The 6 Key Sections of the Report

Once the HTML file is generated and opened, it categorizes the dataset's analysis into six distinct sections:

A. Overview

Provides a high-level summary of the dataset. This includes the total number of variables (columns), total observations (rows), duplicate rows, missing cells, and the total memory footprint. Most importantly, it generates **Warnings** (e.g., "Column X has a high cardinality", "Column Y has 20% missing values") to instantly flag potential data quality issues.

B. Variables (Univariate Analysis)

Breaks down every single column individually.

- **For Numerical data:** Displays the mean, median, min, max, standard deviation, and automatically plots a histogram to show the distribution. It also highlights extreme values (outliers).
- **For Categorical data:** Displays distinct counts, unique category percentages, and plots bar charts for class frequencies.

C. Interactions (Bivariate Analysis)

Provides an interactive grid where you can select any two numerical variables, and it will instantly generate a scatter plot showing their relationship. This eliminates the need to manually write

`sns.scatterplot()` for every combination.

D. Correlations

Automatically calculates and plots heatmaps for different correlation coefficients (like Pearson and Spearman). This allows you to quickly identify multicollinearity (independent variables that are too highly correlated) or strong predictive relationships between features and the target variable.

E. Missing Values

Visualizes missing data through bar charts and matrix plots, helping you spot patterns in the null values (e.g., if data is missing randomly or if it's clustered together systematically).

F. Sample

Displays the `.head()` and `.tail()` of the dataset for quick reference, ensuring the data loaded correctly.

4. Why Automate?

While Pandas Profiling is incredibly powerful, it does *not* replace human intuition. It acts as an accelerator. Instead of spending hours writing plotting code, you generate the report in seconds, allowing you to spend your time **interpreting** the insights and planning your feature engineering strategies.